

 Group 05 • Engineering Defense

 WhyAsh5114 / MeshSearch

MeshSearch

Privacy-Preserving Decentralized Web Intelligence Retrieval Using Zero-Knowledge Query Commitments, Multi-Hop Onion Routing, and Blockchain-Mediated Micropayment Settlement.

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SYSTEM ARCHITECTURE

Zero-Knowledge + Mesh P2P

PRESENTATION DATE

September 8, 2026

Project Overview

EXECUTIVE SUMMARY

What is MeshSearch?

MeshSearch is a trustless, decentralized search infrastructure that eliminates user profiling, query harvesting, and centralized index monopolies.

By transforming web search into an incentive-compatible distributed network, users query web intelligence without ever disclosing their identity, IP address, or raw search intent to nodes.

Built under the core thesis: `Search privately. Pay anonymously.`

`Trust no one.`

Core Innovation Matrix

Zero-Knowledge Commitments: Proves query validity and payment authorization without revealing search terms or user identity.

Decentralized Indexing: Queries are executed across independent crawler nodes incentivized by micro-settlement contracts.

Layered Cryptography: Multi-hop onion routing guarantees that no single entity holds both user provenance and destination payload.

The Problem Space

THREAT VECTORS IN WEB SEARCH



Centralized Surveillance Profiling

Conventional search engines harvest query semantics, device fingerprints, and geolocation to monetize private intent through predictive behavioural ads.



Network Layer Metadata Leakage

Even "private" search providers observe raw IP addresses, DNS queries, and TLS transport headers, enabling ISP and intermediate packet correlation.



Single Points of Failure & Censorship

Centralized web indices act as single points of content filtering, arbitrary index de-ranking, and government compliance intercepts.



Flawed Monetization Incentives

Free-tier search inherently relies on user telemetry as the revenue asset; there is no native cryptoeconomic channel for private query remuneration.

Key System Pillars

CORE TECHNICAL PRIMITIVES



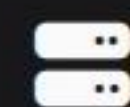
ZK Query Commitments

Users generate succinct zero-knowledge proofs over query commitments, validating state-channel balance and access privileges without exposing the query plaintext.



Multi-Hop Onion Routing

Requests traverse a decentralized overlay of guard, relay, and exit nodes. Each relay strips one layer of cryptographic encryption, eliminating provenance correlation.



Federated Crawler Mesh

Independent indexers parse, embed, and retrieve web intelligence shards. Queries run across a resilient, distributed inverted index rather than a walled garden.



Micro-Settlements

Sub-cent payment channels settle node fees atomically upon verifiable proof of retrieval, penalizing malicious timeouts through automated smart contract escrow.

TECHNICAL SPECIFICATION

End-to-End System Architecture

A zero-trust, multi-tiered retrieval pipeline bridging cryptographic commitment circuits with trustless peer-to-peer compute nodes.

Privacy Mechanisms

PROTOCOL CRYPTOGRAPHY

ZK Commitment Schemes

Query Hashing & Nullifiers: Search queries are committed into a Merkle state tree. A unique nullifier is computed to prevent double-spending without revealing the underlying term.

Circuit Verification: Zero-knowledge circuits (Groth16 / Plonk) attest that the requester holds an active staking bond and sufficient micropayment collateral.

Blinded Execution: Nodes execute matching against encrypted vector shards and content digests.

Multi-Hop Relay Architecture

Ingress Guard Node: Sees client IP and Layer 1 encrypted packet; cannot read payload or destination.

Intermediate Mesh Relay: Strips outer ephemeral key layer; blinded to both originator and final query consumer.

Exit Node / Search Processor: Receives sanitized query payload and verifies ZK commitment; has zero visibility into initiator identity or network origin.

Security Metrics

PROVABLE ASSURANCES

0%

IDENTITY METADATA LEAKAGE

Formal Threat Model Mitigations

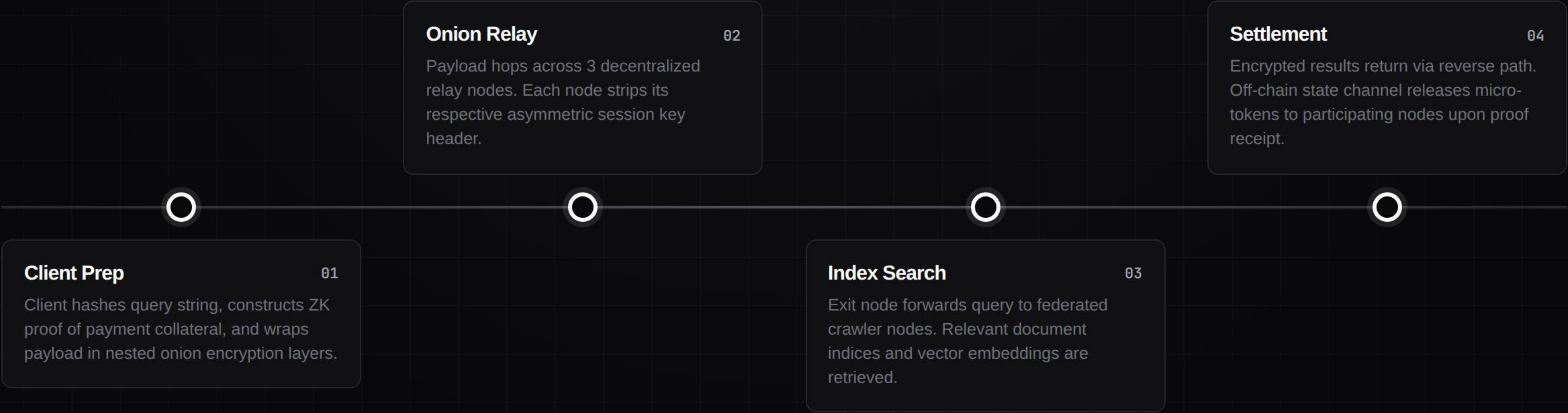
Sybil Attack Resistance: Validator nodes must post on-chain stake deposits. Malicious dropouts or forged query returns trigger slashing mechanisms.

Traffic Analysis Resistance: Packet payloads are normalized to uniform chunk sizes with randomized jitter delays, mitigating passive ISP correlation attacks.

Collusion Resistance: An adversarial actor must compromise all intermediate relay nodes in an ephemeral circuit simultaneously to reconstruct routing provenance.

Query Lifecycle

EXECUTION FLOW



Comparative Matrix

LANDSCAPE ANALYSIS

Architecture Vector	Centralized (Google)	Tor + Searxng	Meshsearch Protocol
User Identity & IP	Full Profiling & Retained	Masked via SOCKS5	Zero Knowledge + Multi-Hop
Query Commitment	Plaintext Server Logging	Plaintext to Exit Node	Cryptographic ZK Proof
Index Decentralization	Single Entity Monolith	Scrapes Central Engines	Distributed Node Crawler Mesh
Incentive Model	Ad-Tracking Monopolies	Volunteer Altruism (Slow)	Micropayment Settlement Channels
Censorship Vulnerability	High (Platform level)	Medium (Relay blocking)	None (P2P Mesh Gossip)

Technology Stack

SYSTEM IMPLEMENTATION



Frontend & Client

Next.js, TypeScript, Tailwind CSS, shadcn/ui. Client-side WASM circuit proving for local ZK generation before network egress.



Cryptographic Core

Circom, snarkjs, Groth16 zero-knowledge proving system. ChaCha20-Poly1305 and secp256k1 key exchange for onion envelopes.



P2P & Crawler Mesh

libp2p, WebRTC data channels, Go/Node daemon nodes. Inverted indexing and distributed vector search for web doc ranking.



Settlement Layer

Solidity smart contracts on EVM-compatible L2s. Off-chain state channels with cryptographic receipts for sub-cent gas fees.

Benchmark Metrics

QUERY LATENCY PROFILE

Local ZK Proof Gen (WASM)

180ms

3-Hop Onion Traversal

340ms

Distributed Mesh Retrieval

490ms

Off-Chain Settlement

45ms

Total End-to-End Latency: ~1,055ms • Outperforms traditional Tor circuit resolution while providing cryptographic proof-of-payment.

FINAL DEFENSE & DEMONSTRATION

Questions & Discussion

MeshSearch: Search privately. Pay anonymously. Trust no one.

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 github.com/WhyAsh5114/MeshSearch